

SBC 201 Chapter 9 Fire Protection Systems — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–18 (design-check rows, not pasted inventory), plus project-use controls, a coverage summary and an unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 9, source file Reference\SBC 201 2024\source_reference\Chapter_09 – FIRE PROTECTION SYSTEMS.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **310** independently checkable numeric records (**251** Verified, **59** Verify source). Unresolved OCR and missing equations are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural/fire-protection coordination advisory matrix. It is not a stamped fire-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 4 (including 403.3), Chapter 10, Chapter 11, Chapter 27, Chapter 30, SBC 501, SBC 701, SBC 801, NFPA 4/10/13/13R/13D/14/20/72, ICC A117.1, ASME A17.1, commentary examples, Figure 903.2, or the existing chapter summary. Where Chapter 9 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications. Chapter 9 uses “high-rise” without restating a height trigger.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. NFPA 13 versus 13R is not assumed as a project choice. **903.3.1.2** eligibility (four stories or fewer and **9 m** above/below fire-department vehicle access) is tested first. For this occupied-floor-above-**23 m** basis, 13R is not a permitted installation path; remaining sprinkler geometry uses the **903.3.1.1 / NFPA 13** path.
4. Mixed-use podium, enclosed parking, amenity assembly, Type I cooking, landscaped roof, fuel-burning appliances and storey/FD-access datums are unconfirmed.
5. Table 906.3(1), Table 906.3(2) Extra-High row, Table 907.5.2.3.2, and the 909.6.2 / 909.10.1 / 910.3.3 equations are concatenated or cut off. Affected values are **Verify source**.
6. Figure 903.2 occupancy-threshold grid is a figure, not a value source. Table 903.2.11.6 names **403.3** as an outbound high-rise suppression pointer; Chapter 4 text is not imported.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, mixed-use/podium condition, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 9 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 9 source text.
Verify source	OCR, flattened table, page-split, missing heading, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 9 application	Required project action
Sprinkler installation standard	Unconfirmed; 13R eligibility is tested, not assumed	903.3.1.2 permits NFPA 13R only at four stories or fewer and highest/lowest floor 9 m of FD vehicle access. High-rise geometry fails that test	Fire engineer to lock 903.3.1.1 / NFPA 13 unless a published 13R path is proven; do not take 13-only trade-offs on a 13R system
Storeys and FD-access datum	Occupied floor stated above 23 m ; exact height, grade plane and vehicle-access level unconfirmed	Controls 13R eligibility, 903.2.11.3 16.5 m / OL 30 , 905.3.1 4 stories / 9 m , 907.2.13.3 36 m multi-channel EVACS, and FSAE sprinkler omission	Issue a signed code datum sheet with grade plane, occupied-floor elevations, basements and lowest FD vehicle access
Mixed-use / podium	Unconfirmed	903.2.8 sprinkles the entire building that contains a Group R fire area. Non-R fire areas may independently require NFPA 13 even if a 13R residential path existed	Freeze occupancy by fire area; do not apply residential 13R omissions to podium retail, parking or assembly
Enclosed / open parking	Unconfirmed	903.2.10 enclosed garage 1115 m² or beneath other groups; open garage 4460 m² ; commercial MV 465 m²	Classify podium/basement parking as 406.5 vs 406.6 and dimension fire areas
Amenity assembly / occupied roof	Unconfirmed	903.2.1 Group A thresholds and 903.2.1.6 occupied-roof OL 100 / 300 can add story-to-LED sprinkler coverage; 907.2.1 alarm thresholds are occupancy-specific	Classify each lounge, gym, pool deck and roof amenity by actual function and occupant load
Type I cooking	Unconfirmed	904.2.2 / 904.13 require an extinguishing system for each required Type I hood (SBC 501 / SBC 801 §609)	Kitchen consultant to lock hood type; if Type I, apply 904.13 geometry
EVACS / FCC location	High-rise stated; AHJ approval unconfirmed	907.2.13 requires EVACS; 911.1.1 location must be approved by the fire code official	Fire alarm designer and SCD consultant to lock FCC location, paging zones and cause-and-effect
Fuel-burning appliances	Unconfirmed all-electric vs gas	915 requires CO detection in Group R where 915.1.2–915.1.6 conditions exist	MEP to state whether any fuel-burning appliance, fireplace, furnace or attached private garage exists
Landscaped roof / helipad	Unconfirmed	905.3.8 extends standpipes to a landscaped roof; 905.3.6 sends heliports to SBC 801	Confirm roof program before freeze of standpipe termination

Decision / gap	Current project basis	Why it controls Chapter 9 application	Required project action
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	Acceptance of NFPA 13 design, smokeproof method, FCC and radio coverage cannot be concluded from Chapter 9 alone	Engage the qualified local/fire consultant before design freeze

4. Scope, required systems and high-rise testing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
901.1	Chapter scope	This chapter specifies where fire protection and life safety systems are required and governs their design, installation and operation	All buildings in scope	Maintenance/testing live in SBC 801 (not imported)	Direct	Treat Chapter 9 as the charging chapter for sprinklers, standpipes, alarms, smoke control, FCC, FDC and fire pumps	Verified
901.2	Required-system status	Install, repair, operate and maintain systems per this code and SBC 801. Any system used to take an exception or reduction is a required system	Fire protection or life safety system	Voluntary systems still must meet this chapter	Direct	If a sprinkler or alarm trade-off is taken elsewhere, keep that system as required	Verified
901.5	Acceptance before occupancy	Do not occupy a portion of a structure until the required fire protection systems in that portion have been tested and approved	New work / partial occupancy	Tests in the presence of the building official where required	Direct	Sequence partial occupancy only after the serving systems are tested	Verified
901.6 / 901.6.1	Supervising-station monitoring	Where required, monitor fire protection systems at an approved supervising station per NFPA 72. Automatic sprinkler systems shall be so monitored	Required sprinkler systems	Exception 1: one- and two-family dwellings. Exception 2: limited-area systems per 903.3.8	Direct	Show supervising-station monitoring on the sprinkler/alarm riser diagram	External verification
901.6.2.1	High-rise integrated testing	Integrated testing shall comply with NFPA 4, with a test before certificate of occupancy and at intervals not exceeding 10 years , unless an NFPA 4 integrated-system test plan specifies otherwise	High-rise buildings	Repeat of the entire integrated test is not required after a repair except to verify functions initiated by replaced equipment	Direct	Put NFPA 4 integrated testing on the commissioning specification; do not import NFPA 4 procedures here	External verification
901.6.2.2	Smoke-control integrated testing	Where a fire alarm is integrated with a Section 909 smoke control system, integrated testing shall comply with NFPA 4 on the same 10-year cadence	Fire alarm integrated with 909 smoke control	Same repair-repeat limitation as 901.6.2.1	Conditional	If stair pressurization or another 909 system is provided, include it in the NFPA 4 test plan	External verification
901.6.3	Fire-alarm supervising station	Fire alarm systems required by 907.2 shall be monitored by an approved supervising station per 907.6.6	Required 907.2 fire alarm	Exceptions: 907.2.11 smoke alarms; Group I-3 smoke detectors; one- and two-family sprinklers	Direct	Connect the tower fire alarm to a supervising station; keep unit smoke alarms off that path	Verified
901.7	Fire-area separation	To stay under a Chapter 9 protection threshold, divide fire areas with 706 fire walls, 707 fire barriers or 711 horizontal assemblies rated not less than Section 707.3.10	Building subdivided to avoid a protection trigger	Rating values live in 707.3.10 (not imported)	Conditional	903.2.8 already sprinkles any building with a Group R fire area; do not use fire areas to omit tower sprinklers	External verification

5. Fire pump and riser rooms

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
902.1	Pump/riser room working space	Size rooms for manufacturer clearances so equipment can be inspected, serviced, repaired or replaced without removing permanent construction or disabling a required fire-resistance-rated assembly; provide a door and unobstructed passageway for the largest piece of equipment	Fire pump room or sprinkler riser room is provided	This section does not itself require a room	Direct	Coordinate room size and a removal path with the fire-protection contractor before the core is frozen	Verified
902.1.1	Pump/riser access	Provide ready access to risers, fire pumps and controllers. A locked room door is permitted if the key is available at all times	Equipment in a pump or riser room	None stated	Direct	Show a 24-hour key location on the life-safety plan	Verified
902.1.2	Access-door lettering	Label access doors with an approved contrasting sign; letters minimum height 50 mm , minimum stroke 10 mm	Pump or riser room doors	None stated	Direct	Put 50 mm / 10 mm lettering on the door schedule	Verified
902.1.3	Pump/riser room temperature	Maintain the room at not less than 4 °C ; heating units shall be permanently installed	Pump or riser room	None stated	Direct	Specify permanent heating to 4 °C in the pump/riser room	Verified
902.1.4	Pump/riser room lighting	Provide permanently installed artificial illumination	Pump or riser room	None stated	Direct	Put permanent lighting on the electrical room data sheet	Verified

6. Group R sprinkler charging

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.2	Where sprinklers are required	Provide approved automatic sprinklers in new buildings in the locations in 903.2.1 through 903.2.12	New buildings	Telecom equipment spaces with 907.2 smoke detection and 1-hour barriers or 2-hour horizontal assemblies, or both	Direct	Do not use the telecom exception for R-2 dwelling floors	Verified
903.2.8	Group R building sprinklers	Provide an automatic sprinkler system per 903.3 throughout all buildings with a Group R fire area	Any Group R fire area, including R-2	903.2.8.1–903.2.8.4 permit 13D/13R only for R-3, R-4 and small care dwellings — not this R-2 high-rise	Direct	Sprinkle the entire building that contains the R-2 fire area, including mixed-use stories	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.2.11.3	Height 16.5 m sprinkler trigger	Sprinkle throughout buildings that have one or more stories with occupant load 30 or more located 16.5 m or more above the lowest level of fire department vehicle access, measured to the finished floor	Occupied story at that height and load	Group F-2 occupancies	Direct	Treat as a second independent trigger; 903.2.8 already covers this tower	Verified
Table 903.2.11.6	Outbound high-rise suppression pointer	Additional required suppression systems include 403.3 High-rise buildings among other outbound sections	Uses listed in the table	Concatenated OCR pointer table; no Chapter 4 value is adopted	External verification	Obtain published 403.3; do not import high-rise sprinkler text from Chapter 4 here	Verify source

7. Sprinkler installation standard (13 vs 13R)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.3.1	Default installation standard	Design and install sprinklers per 903.3.1.1 unless 903.3.1.2, 903.3.1.3 or another chapter permits otherwise	Required automatic sprinklers	13R and 13D only where those sections are met	Direct	Start from NFPA 13 unless a numbered 13R/13D permission is proven	External verification
903.3.1.2	NFPA 13R eligibility	NFPA 13R is permitted in Group R only where all apply: four stories or fewer above grade plane; highest floor 9 m or less above lowest FD vehicle access; lowest floor 9 m or less below that access. Stories under 510.2 / 510.4 are measured from grade plane	Group R proposing 13R	Pedestal 510.2 / 510.4 does not restart the four-story / 9 m tests from the podium roof	Direct	Record that this occupied-floor-above- 23 m high-rise fails 13R eligibility; do not specify 13R	Verified
903.3.1.1	NFPA 13 throughout	Where the code requires a building or portion to be equipped throughout per this section, install sprinklers throughout per NFPA 13 except 903.3.1.1.1 and 903.3.1.1.2	13R/13D not permitted	Exempt locations and R bathrooms as numbered	Direct	Specify NFPA 13 for the tower; installation details live in NFPA 13 (not imported)	External verification
903.3.1.3	NFPA 13D scope	NFPA 13D is permitted in one- and two-family dwellings, Group R-3, Group R-4 Condition 1 and townhouses	Those occupancies only	Not a Group R-2 high-rise path	Not typical	Omit 13D from the tower specification	Verified
903.3.2	Quick-response / residential sprinklers	Where sprinklers are required, install quick-response or residential sprinklers in dwelling units and sleeping units in Group R occupancies (and other listed I-2/ambulatory/light-hazard locations)	R dwelling/sleeping units	Listings and 903.3.1 still apply	Direct	Specify QR or residential heads in units; keep other occupancies on their NFPA 13 hazard	Verified

8. Sprinkler omissions, obstructions and high-rise floor valves

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.3.1.1.1	NFPA 13 exempt rooms	Sprinklers may be omitted in listed rooms if an approved 907.2 automatic fire detection system is provided. Do not omit sprinklers merely because a room is damp, rated or contains electrical equipment. Item 3: generator/transformer rooms with 2-hour walls and floor/ceiling or roof/ceiling assemblies. Items 5–6: fire-service-access and occupant-evacuation elevator machine/control rooms	Designer proposes an omission	FSAE/OEE rooms are Conditional on Chapters 4/30 requiring those elevators; those chapters are not imported	Conditional	Use omissions only with listed detection and the 2-hour generator-room rating; keep unit and corridor sprinklers	Verified
903.3.1.1.2	Small R bathroom omission	In Group R, sprinklers are not required in bathrooms not exceeding 5 m² in individual dwelling or sleeping units if walls and ceilings (including behind shower/tub) are noncombustible or limited-combustible with a 15-minute thermal barrier	Unit bathrooms	Accessible bathrooms often exceed 5 m² — then sprinkle	Conditional	Dimension each unit bathroom; sprinkle any bathroom over 5 m²	Verified
903.2.11.2	Rubbish and linen chutes	Install sprinklers at the top of rubbish and linen chutes and in terminal rooms; additional heads at alternate floors and at the lowest intake. Recess and freeze-protect extension sprinklers; provide access for servicing	Rubbish or linen chutes	Heading is missing in the extract; body text follows 903.2.11.1.3	Direct	Show chute sprinklers at top, terminal, lowest intake and alternate floors on the riser diagram	Verify source
903.3.3	Obstruction and pile clearance	Install sprinklers with regard to obstructions per the applicable standard. Sprinkle in or under covered kiosks, displays, booths, concession stands or equipment exceeding 1200 mm in width. Maintain not less than 900 mm clearance between sprinklers and the top of combustible-fiber piles	Obstructions or combustible-fiber storage	Kitchen equipment under a 904 exhaust hood is excepted	Conditional	Keep 1200 mm kiosks sprinkled and 900 mm above fiber storage in amenities/back-of-house	Verified
903.3.8.1	Limited-area sprinkler cap	Limited-area systems shall not exceed six sprinklers in any single fire area	Limited-area system proposed	Not a substitute for 903.2.8 throughout-building protection	Not typical	Do not use a six-head limited-area system to satisfy the R-2 building trigger	Verified
903.4	Electrical supervision	Electrically supervise valves, pumps, tanks, water levels and temperatures, critical air pressures and waterflow switches by a listed fire alarm control unit	Automatic sprinkler systems	Eight listed exceptions (13D dwellings, limited-area, some 13R combination supplies, sealed jockey/kitchen/fuel/trim valves, underground roadway valves)	Direct	Supervise floor and supply valves at the FACU; do not apply the 13R combination-supply exception to this 13 tower	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.4.2	Exterior waterflow alarm	Connect an approved exterior audible device, activated by flow equivalent to one smallest-orifice sprinkler. Where a fire alarm is installed, sprinkler actuation shall actuate the building fire alarm	Each automatic sprinkler system	None stated	Direct	Show the exterior waterflow bell and fire-alarm interconnection	Verified
903.4.3	High-rise floor control valves	Provide approved supervised indicating control valves at the point of connection to the riser on each floor in high-rise buildings	High-rise buildings	None stated	Direct	Put a supervised floor control valve at each story on the combined standpipe/sprinkler riser	Verified

9. Commercial cooking and alternative extinguishing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
903.1.1 / 904.2.1	Alternative system vs sprinkler trade-offs	A 904 alternative extinguishing system may replace occupancy sprinklers where recognized and approved, but it shall not unlock sprinkler exceptions or reductions	Alternative agent proposed in lieu of 903	Does not make the building “sprinklered throughout” for other chapters	Conditional	Do not take egress or rating reductions on a clean-agent or hood system	Verified
904.2.2	Type I hood extinguishing	Each required commercial kitchen exhaust hood and duct system required by SBC 801 §609 or SBC 501 Chapter 5 to have a Type I hood shall be protected with an approved automatic fire-extinguishing system	Type I hood required	Hood-type rules live in SBC 501 / SBC 801 (not imported)	Conditional	If a Type I amenity or staff kitchen exists, specify a 904.13 system	External verification
904.13.1	Hood manual actuator	Locate a manual actuation device at or near a cooking-area means of egress, not less than 3 m and not more than 6 m from the kitchen exhaust system; install not more than 1200 mm and not less than 1050 mm above the floor; maximum force 178 N and maximum movement 350 mm	904.13 commercial cooking system	Automatic sprinkler systems need not have this manual means	Conditional	Dimension the pull station on the kitchen plan	Verified
904.13.2	Cooking fuel/power shutdown	Actuation shall automatically shut down fuel or electrical power to the cooking equipment; reset shall be manual	904.13 system	None stated	Conditional	Show interlock to gas/electric cooking equipment	Verified
904.14	Domestic cooking in R-2 dormitories	Cooktops and ranges in Group R-2 college dormitories per 420.11 shall be protected per 904.14.1 (UL 300A hood system or 30-minute ignition-resistant burners)	R-2 college dormitory domestic cooking	Ordinary R-2 apartments are not this trigger	Conditional	Open only if the building is college/university staff or student housing with 420.11 cooking	External verification

10. Standpipe systems

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
905.2	Standpipe standard	Install standpipes per this section and NFPA 14; FDCs per 912	Required standpipe	Hydraulic criteria live in NFPA 14 (not imported)	Direct	Specify NFPA 14; do not import residual-pressure values from memory	External verification
905.3.1	Height-based Class III standpipe	Install Class III standpipes throughout where any apply: four or more stories above or below grade plane; highest floor more than 9 m above lowest FD vehicle access; lowest floor more than 9 m below highest FD vehicle access	Building height vs FD access	Exception 1: Class I allowed if sprinklered per 903.3.1.1 or 903.3.1.2. Exceptions 2–5: Class I in B/E, parking, sprinkled basements, or where occupant hose will not be used. Exception 6: ignore recessed docks ≤ four vehicles and impractical topography	Direct	Provide at least Class I (sprinklered-building exception) on a combined riser; lock wet vs manual with NFPA 14 and the high-rise status	Verified
905.3.8	Landscaped-roof standpipe	If the building has a standpipe and a landscaped roof, extend the standpipe to that roof level	Landscaped roof present	None stated	Conditional	Terminate a hose valve at the landscaped roof	Verified
905.4	Class I hose-connection locations	Provide Class I hose connections: in every required interior exit stairway at each story (main floor landing unless otherwise approved); each side of a horizontal-exit wall; every exit passageway entrance; roof with slope less than 4:12 (33.3-percent) at the roof or highest stair landing with 1011.12 access; additional locations if remote floor portion exceeds 45 m (nonsprinklered) or 60 m (sprinklered) from a hose connection	Class I standpipe	Single connection permitted in an open corridor/breezeway between open stairs not greater than 23 m apart. Horizontal-exit / exit-passageway valves may be omitted if the floor is reachable by a 9 m hose stream from a nozzle on 30 m of hose	Direct	Place a hose valve at each stair floor landing; add extra valves if travel exceeds 60 m on this sprinkled tower	Verified
905.4.1	Standpipe riser protection	Protect Class I risers and laterals not in an interior exit stairway to the same fire resistance as required vertical enclosures	Combined riser outside a stair	In a fully sprinkled building, laterals outside a stair need not be in rated construction	Direct	Keep the main riser in the exit stair; if a lateral leaves the stair, apply the sprinkled-building lateral exception or match shaft rating	Verified
905.4.2	Interconnection	Where more than one standpipe is provided, interconnect per NFPA 14	Multiple standpipes	None stated	Direct	Interconnect risers; do not import NFPA 14 supply sizes here	External verification
905.7.1	Cabinet lettering	Identify cabinets with a permanently attached contrasting sign, letters not less than 50 mm high	Fire-fighting equipment cabinets	Pictogram if the door is too small; glass-panel doors need not be marked	Conditional	If hose/extinguisher cabinets are used, apply 50 mm lettering	Verified

11. Portable fire extinguishers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
906.1	R-2 extinguisher locations	Install portable extinguishers in Group R-2 occupancies. Exception 1: if each dwelling unit has a minimum 1-A:10-B:C extinguisher, common-area extinguishers are required only in Items 2–6 (cooking 9 m , flammable liquids, construction, Table 906.1, special-hazard rooms)	Group R-2	Item 2 cooking 9 m also names R-2 college dormitories separately	Direct	Either distribute NFPA 10 extinguishers in common areas or put a 1-A:10-B:C unit in every dwelling and still cover Items 2–6	Verified
906.3.1 / Table 906.3(1)	Class A distribution	Size and distribute Class A extinguishers per Table 906.3(1). Concatenated OCR: Light/Ordinary/Extra minimum ratings, 280 / 140 / 100 m² per unit of A, 1,045 m² max area, travel 23 m . No reconstructed cell is adopted	Class A hazard occupancy	Footnote a: two 9.5 Liter water-type = one 4-A. Footnote c: two 1-A water-type = one 2-A in Light Hazard	Direct	Verify the published table before locking rating and 23 m travel; do not copy flattened 280 / 140 / 100 tokens as design-release values	Verify source
906.3.2 / Table 906.3(2)	Class B shallow-liquid distribution	For flammable/combustible liquids ≤ 6.5 mm deep, use Table 906.3(2). Extra-High row is concatenated 20-B1540-B980-B15 . No reconstructed rating/travel pair is adopted	Class B hazard with shallow liquid	Deeper liquids go to NFPA 10 (not imported)	Conditional	If a shallow Class B hazard exists, verify the published Extra-High row before specifying 20-B / 40-B / 80-B travel	Verify source
906.9.1–906.9.3	Extinguisher mounting	Gross weight ≤ 18 kg : top not more than 1500 mm above the floor. Gross weight > 18 kg : top not more than 1000 mm . Bottom clearance not less than 100 mm	Hand-held extinguishers	Wheeled units follow 906.10	Direct	Put mounting heights on the interior-detail sheet	Verified

12. Group R-2 fire alarm and unit smoke alarms

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
907.2.9.1	R-2 manual fire alarm	Install a manual fire alarm that activates 907.5 occupant notification where any apply: a dwelling/sleeping unit is three or more stories above the lowest LED; a unit is more than one story below the highest LED of its exits; the building contains more than 16 dwelling or sleeping units	Group R-2	Exception 1: ≤ two stories with 1-hour partitions and each unit exiting directly to a public way/court/yard. Exception 2: omit manual boxes if sprinkled per 903.3.1.1 or 903.3.1.2 and waterflow activates occupant notification throughout the zones. Exception 3: omit the fire alarm in sprinkled buildings with no interior corridors and direct/open-ended-corridor egress	Direct	This high-rise trips the three-story and likely the 16-unit tests. Exception 2 may omit pull stations but does not delete the 907.2.13 high-rise alarm/EVACS	Verified
907.2.9.2 / 907.2.11.2	Unit smoke-alarm locations	Install listed UL 217 single- or multiple-station smoke alarms regardless of occupant load: on the ceiling or wall outside each separate sleeping area in the immediate vicinity of bedrooms; in each sleeping room; on each story within the dwelling unit including basements (not crawl spaces or uninhabitable attics). Split levels without an intervening door may share the upper-level alarm if the lower level is less than one full story below	R-2 dwelling/sleeping units	907.2.11.7 fire-alarm smoke detectors may substitute with in-unit-only notification	Direct	Show three-location smoke alarms on every unit typical	Verified
907.2.11.3	Smoke alarms near cooking	Do not install ionization alarms less than 6 m horizontally from a permanent cooking appliance; ionization with a silence switch less than 3 m ; photoelectric less than 1.8 m , unless that would prevent a required 907.2.11.2 location	Cooking appliance in or adjacent to a required alarm location	Required-location override is stated	Direct	Dimension kitchen/living smoke alarms off the cooktop	Verified
907.2.11.4	Smoke alarms near bathrooms	Install smoke alarms not less than 900 mm horizontally from a bathroom door/opening with a bathtub or shower unless that would prevent a required location	Bathroom with tub/shower	Required-location override is stated	Direct	Keep bedroom alarms 900 mm from the bath door	Verified
907.2.11.5	In-unit interconnection	Where more than one smoke alarm is required in a unit, activation of one shall activate all in that unit. Listed wireless interconnection is permitted. The alarm shall be clearly audible in all bedrooms with intervening doors closed	Group R units	Physical wiring not required for listed wireless	Direct	Specify interconnect (wired or listed wireless) on the unit electrical typical	Verified
907.2.11.6	Smoke-alarm power	Primary power from building wiring on a commercial source, with battery backup. Integral-strobe alarms without battery backup shall use Section 2702 emergency power. Permanent wiring; no disconnect other than overcurrent protection	New construction	Battery backup not required if connected to a 2702 emergency system	Direct	Circuit unit smoke alarms from house power with battery backup	External verification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
907.2.9.3	College/university R-2 detection	Automatic smoke detection activating 907.5 is required in college/university student or staff housing in common spaces, laundry/mechanical/storage rooms, and interior corridors serving units, with unit smoke alarms interconnected to the fire alarm per NFPA 72	R-2 operated by a college or university for student or staff housing	Exception: no interior corridors and each unit opens to exterior exit access or directly to an exit	Conditional	Open only if the occupancy is campus housing; do not apply to a market apartment tower	External verification

13. High-rise detection, EVACS and fire-department communications

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
907.2.13	High-rise alarm package	Provide an automatic smoke detection system per 907.2.13.1, a fire-department communication system per 907.2.13.2, and an emergency voice/alarm communication system per 907.5.2.2	High-rise buildings	Airport towers, open parking, A-5, low-hazard special occupancies, H-1/H-2/H-3; I-1/I-2 staff-attended private mode	Direct	Include detection, FD communications and EVACS on the fire-alarm riser; R-2 is not a listed exception	Verified
907.2.13.1.1	High-rise area smoke detection	Connect smoke detectors to the automatic fire alarm; activation shall start EVACS. In addition to 907.2.1–907.2.9: detectors in each unsprinklered mechanical/electrical/transformer/telephone or similar room, and in each elevator machine room, machinery space, control room, control space and elevator lobbies	High-rise	Elevator rooms and lobbies require detectors even if sprinkled	Direct	Show lobby and elevator-machine-room detectors on every typical floor	Verified
907.2.13.1.2	High-rise duct smoke detection	Duct detectors per 907.3.1: in the main return/exhaust plenum of each air-conditioning system greater than 940 liters per second , downstream of the last inlet; and at each connection to a vertical duct/riser serving two or more stories. In Group R-2, a smoke detector is allowed in each return-air riser carrying not more than 2360 liters per second and serving not more than 10 air-inlet openings	HVAC capacity and risers	R-2 riser alternative is optional	Direct	Coordinate duct detectors with mechanical; use the R-2 2360 L/s / 10-inlet alternative only where those limits are met	Verified
907.2.13.2	Wired FD communication alternative	Where a wired communication system is approved in lieu of in-building two-way emergency-responder coverage per SBC 801 §510, install it per NFPA 72 between the 911 fire command center, elevators, elevator lobbies, emergency/standby power rooms, fire-pump rooms, areas of refuge and interior exit stairways, with a device at each stair floor	Wired system approved in lieu of radio coverage	Default path is SBC 801 §510 radio coverage (918.1)	Conditional	Prefer 918.1 radio coverage; if a wired substitute is approved, put jacks at the listed rooms and each stair floor	External verification
907.2.13.3	Multi-channel voice evacuation	In buildings with an occupied floor more than 36 m above the lowest level of fire department vehicle access, high-rise voice evacuation shall be multiple-channel	Occupied floor > 36 m above FD access	Occupied floor stated above 23 m but 36 m is unconfirmed	Conditional	If the occupied floor exceeds 36 m , specify a multi-channel EVACS	Verified
907.5.2.2	EVACS operation and paging zones	Design EVACS per NFPA 72. Automatic detector, waterflow or manual box shall sound an alert tone then voice instructions. In high-rise buildings the system shall operate on at least the alarming floor, the floor above and the floor below. Provide speakers throughout by paging zones, minimum: elevator groups; interior exit stairways; each floor; areas of refuge	Required EVACS	Group I-1/I-2 constantly attended / overhead page	Direct	Zone EVACS by floor ±1, stairs, elevator groups and any area of refuge	External verification
907.5.2.2.1–907.5.2.2.3	EVACS override and live voice	Provide manual override on a selective and all-call basis; live voice by paging zone; other announcements allowed if fire-alarm use takes precedence	EVACS	None stated	Direct	Show selective/all-call microphones at the FCC	Verified
907.5.2.2.5	EVACS standby power	Provide standby power per Section 2702	EVACS	2702 values are not imported	Direct	Put EVACS on the standby-power schedule	External verification

14. Occupant notification, zoning and R-2 visible capability

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
907.4.2.1	Manual-box location	Locate manual fire alarm boxes not more than 1500 mm from each exit entrance. In buildings not sprinkled per 903.3.1.1 or 903.3.1.2, additional boxes so travel does not exceed 60 m	Manual fire alarm required	907.2.9.1 Exception 2 may omit boxes if waterflow notifies	Direct	If pull stations remain, place them within 1500 mm of each exit; the 60 m extra-box rule does not apply to this 13-sprinkled tower	Verified
907.4.2.2	Manual-box height	Activating handle/lever not less than 1000 mm and not more than 1200 mm above the floor	Manual boxes provided	None stated	Conditional	Put 1000–1200 mm on the device height typical	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
907.5.2.1.1–907.5.2.1.2	Audible sound pressure	Provide 15 dBA above average ambient or 5 dBA above the maximum sound lasting not less than 60 seconds , whichever is greater, in every occupiable space. Combined level shall not exceed 110 dBA ; if average ambient exceeds 105 dBA , visible appliances per NFPA 72 replace audible	Occupant notification	I-2 critical-care exceptions are not this occupancy	Direct	Have the fire-alarm designer prove 15 / 5 / 110 dBA in units, corridors and amenities	Verified
907.5.2.1.3.1–907.5.2.1.3.2	520 Hz sleeping-room signals	In R-2 sleeping rooms, the fire-alarm audible signal shall be a 520-Hz low-frequency signal per NFPA 72. Where 907.2.9 requires a fire alarm, unit smoke-alarm signals in sleeping rooms shall also be 520 Hz , or a listed 520 Hz appliance/detector shall provide it	R-2 sleeping rooms	Applies because 907.2.9 / 907.2.13 require a fire alarm	Direct	Specify 520 Hz bases or sounders in bedrooms	External verification
907.5.2.3.1	Visible alarms in common areas	Provide visible alarm notification appliances in public-use and common-use areas	Required fire alarm	Employee work areas with audible coverage: circuits initially designed with not less than 20-percent spare capacity for future visibles	Direct	Put strobes in lobbies, corridors, amenities and other common-use rooms	Verified
907.5.2.3.3	R-2 future visible capability	In Group R-2 required by Section 907 to have a fire alarm, each story containing dwelling or sleeping units shall support future visible appliances per ICC A117.1 Chapter 11 (wired or wireless)	R-2 with a required fire alarm	A117.1 geometry is not imported	Direct	Show spare capacity / access points per 907.5.2.3.3.1; obtain A117.1 Chapter 11 separately	External verification
907.5.2.3.3.1	Wired future-visible capacity	Wired path: replacement with combination appliances, or future extension from unit smoke-alarm locations, or fire-alarm power supply and circuits with not less than 5-percent excess capacity and a single access point on every story (circuits need not extend beyond that point). Document on 907.1.2 shop drawings	Wired future-visible option	Wireless equipment is an alternative under 907.5.2.3.3	Conditional	If wired, show 5-percent spare and one access point per residential story	Verified
907.6.3	Initiating-device identification	Identify specific device address, location, type, floor and status (normal/alarm/trouble/supervisory)	Fire alarm system	Exceptions include single-story buildings < 2100 m² , systems with only pull stations + waterflow + not more than 10 additional devices, devices that cannot support IDs, and replacements	Direct	Specify addressable identification; the small-building exceptions do not fit this high-rise	Verified
907.6.4	Alarm zoning	Zone each floor separately; a zone shall not exceed 2100 m² and shall not exceed 90 m in any direction	Fire alarm system	Sprinkler zones may follow NFPA 13 area limits	Direct	Split large floor plates so no alarm zone exceeds 2100 m² / 90 m	Verified
907.6.4.2	High-rise separate initiating zones	In high-rise buildings, provide a separate zone by floor for smoke detectors, sprinkler waterflow devices, manual boxes and other approved automatic fire-protection systems, where provided	High-rise	None stated	Direct	Annunciate detectors, waterflow and pull stations as separate floor zones at the FCC	Verified
907.6.6	Fire-alarm monitoring	Required fire-alarm systems shall be monitored (source text is OCR-repeated). Exceptions in the remnant include Group I-3 smoke detectors and one- and two-family sprinklers	Required fire alarm	Confirm published 907.6.6 wording	Direct	Monitor the system at a supervising station per 901.6.3; verify published 907.6.6 before specification freeze	Verify source

15. Smokeproof enclosures (Chapter 9 geometry)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
909.20	Smokeproof-enclosure trigger	Where required by Section 1023.12, construct a smokeproof enclosure per this section (stair/ramp plus open balcony, ventilated vestibule, or pressurized stair and vestibule). Roof access required by SBC 801 shall be from the smokeproof enclosure	1023.12 requires a smokeproof enclosure	1023.12 values are not imported	External verification	Confirm Chapter 10 whether high-rise interior exit stairs must be smokeproof; then pick one 909.20 method	External verification
909.20.1	Vestibule size	Access by vestibule or open exterior balcony. Vestibule minimum dimension not less than the required corridor width, and not less than 1.1 m wide and 1.8 m long in the direction of egress travel	Vestibule method	Pressurization alternative 909.20.5 may omit the vestibule	Conditional	If a vestibule is used, plan 1.1 m × 1.8 m clear	Verified
909.20.3.2–909.20.3.3	Natural-ventilation vestibule	Vestibule-to-stair door not less than a 20-minute fire-protection rating. Each vestibule: minimum net 1.5 m² opening facing an outer court, yard or public way not less than 6 m wide	Natural-ventilation alternative	Building-to-vestibule door is a Section 716 fire door	Conditional	Use only where an outdoor court ≥ 6 m can take the 1.5 m² opening	Verified
909.20.4.2	Mechanical-vestibule ventilation	Not less than one air change per minute; exhaust not less than 150 percent of supply; supply within 150 mm of the floor; exhaust register top not more than 150 mm down from the top of the smoke trap	Mechanical-ventilation alternative	Engineered path 909.20.4.2.1: 90 air changes per hour, sized for three vestibules	Conditional	If this method is selected, show supply/exhaust registers and the smoke trap on the stair detail	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
909.20.4.3	Vestibule smoke trap	Vestibule ceiling not less than 500 mm higher than the door opening into the vestibule	Mechanical-ventilation alternative	Height may be decreased if approved and justified by design and test	Conditional	Keep a 500 mm ceiling step at the vestibule door	Verified
909.20.4.4	Shaft pressure vs vestibule	Maintain minimum 25 Pa positive in the shaft relative to the vestibule with all doors closed	Mechanical-ventilation alternative	None stated	Conditional	Commission 25 Pa shaft-to-vestibule	Verified
909.20.5	Stair pressurization without vestibule	Where sprinkled throughout per 903.3.1.1, the vestibule is not required if each interior exit stair/ramp is pressurized to not less than 25 Pa and not more than 85 Pa relative to the building, all doors closed, under maximum stack and wind	NFPA 13 building electing pressurization-only	This is the usual high-rise 13 path	Conditional	If vestibules are omitted, design 25–85 Pa stair pressurization	Verified
909.20.6.2–909.20.6.3	Pressurized stair and vestibule	Stair $\geq$ 12.50 Pa positive vs vestibule; vestibule $\geq$ 12.50 Pa positive vs the fire floor; door-opening force not more than 133 N . Relief vent not less than 1180 liters per second at the design pressure difference	Pressurized stair + vestibule alternative	Door assemblies as 909.20.6.1	Conditional	If this method is selected, meet the two 12.50 Pa differentials and the 1180 L/s relief	Verified
909.20.7.1	Independent ventilation protection	Smokeproof ventilation shall be independent of other building ventilation. Equipment, control wiring, power wiring and ductwork: exterior; or inside the enclosure; or inside the building behind 2-hour barriers/horizontal assemblies	Mechanical 909.20.4/5/6 methods	Wiring outside the 2-hour enclosure: UL 2196 2-hour cable, 50 mm concrete encasement, or 2-hour electrical circuit protective system	Conditional	Keep pressurization plant on a 2-hour path or listed 2-hour wiring	Verified
909.20.7.2	Smokeproof standby power	Mechanical vestibule/shaft ventilation and automatic fire detection shall have standby power per Section 2702	Mechanical smokeproof methods	2702 values are not imported	Conditional	Put smokeproof fans and detectors on standby power	External verification
909.21.1	Elevator hoistway pressurization	Where hoistway pressurization is used in lieu of required enclosed elevator lobbies: 25 Pa minimum and 65 Pa maximum vs adjacent occupied space, measured at mid-door with cars at recall. Outside air intake not less than 6 m from any exhaust	Pressurization in lieu of 3006 lobbies	Exception 1: on floors containing only Group R, measure vs a dwelling/sleeping unit. FSAE/OEE enclosed lobbies are not replaced by pressurization (named in commentary — not used as a value)	Conditional	Use only if Chapter 30 allows this substitute; commission 25–65 Pa and a 6 m intake offset	Verified

General Section 909 atrium/assembly smoke-control equations, 910 smoke-and-heat vents (Group F-1/S-1 **4650 m²**), and Group I corridor-door rules are omitted from project-use unless mixed-use opens those features in the gap register.

16. Fire command center

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
911.1	Fire command center required	Provide a fire command center in buildings classified as high-rise, and in F-1/S-1 occupancies with footprint over 46450 m²	High-rise classification	F-1/S-1 footprint trigger is not this occupancy	Direct	Provide an FCC; do not use the 9 m² F-1/S-1 size	Verified
911.1.1	FCC location	Location and accessibility shall be approved by the fire code official	Required FCC	None stated	Direct	Obtain fire-code-official approval of FCC location, typically at the main fire-department entry	Verified
911.1.2	FCC separation	Separate from the remainder of the building by not less than a 1-hour fire barrier (707) or horizontal assembly (711), or both	Required FCC	None stated	Direct	Draw a 1-hour FCC enclosure with listed opening protectives	Verified
911.1.3	FCC size	Not less than 0.015 percent of the total building area served or 20 m² , whichever is greater, with minimum dimension 0.7 times the square root of the room area or 3 m , whichever is greater	High-rise FCC	F-1/S-1 footprint > 46500 m² may use 9 m² / 2.4 m where approved — not this tower. Do not use commentary 18.35 m²	Direct	Plan the FCC at least 20 m² with minimum dimension 3 m , and increase if 0.015 percent of building area is greater	Verified
911.1.5	FCC storage ban	Storage unrelated to FCC operation is prohibited	Required FCC	None stated	Direct	Keep the FCC off the furniture/storage inventory	Verified
911.1.6	FCC required features	Comply with NFPA 72 and provide the listed controls and information: EVACS control unit; FD communications; fire-alarm annunciator; elevator status; air-distribution status/controls; 909.16 firefighter's smoke-control panel where smoke control is installed; simultaneous interior-exit-stair unlock; sprinkler valve/waterflow display; emergency/standby power status; FD telephone; fire-pump status; schematic plans; Building Information Card (13.1–13.7); work table; generator supervision/start/transfer; public-address where required elsewhere; elevator fire-recall switch (ASME A17.1); elevator emergency/standby power selector where provided	Required FCC	NFPA 72 layout and ASME A17.1 switch details are not imported	Direct	Freeze an approved 911.1.4 layout that includes every listed feature; do not treat the work table as optional	External verification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
911.1.7	FCC identification	Permanent easily visible sign on the door reading FIRE COMMAND CENTER	Required FCC	None stated	Direct	Put the exact legend on the door schedule	Verified

17. Fire department connections, fire pumps and shaft markings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
912.2–912.2.1	FDC location	Locate FDCs so connected apparatus and hose will not obstruct access for other apparatus; location approved by the fire code official. Place on the street side or facing approved fire-apparatus access roads, fully visible and recognizable, unless otherwise approved	Sprinkler and standpipe FDCs	Existing-building sign path is 912.2.2	Direct	Coordinate FDC location with SCD access; keep it visible from the approach road	Verified
912.2.2	Existing FDC sign	Where an existing FDC is not visible, provide an approved sign with letters FDC not less than 150 mm high and other words or an arrow not less than 50 mm high	Existing building, FDC not visible	New-build FDCs should be visible per 912.2.1	Conditional	Use only on existing-building alterations	Verified
912.4.2	FDC working space	Maintain not less than 900 mm width, 900 mm depth and 1.95 m height in front of and to the sides of wall-mounted FDCs and around free-standing FDCs, unless otherwise required or approved	Wall-mounted or free-standing FDC	Fire-code-official may modify	Direct	Keep a 900 × 900 × 1.95 m clear box at the FDC	Verified
912.5	FDC identification sign	Metal sign with raised letters not less than 25 mm : AUTOMATIC SPRINKLERS , STANDPIPES , TEST CONNECTION , or a combination; identify served portions if the FDC does not serve the entire building	Sprinkler, standpipe or fire-pump FDC	None stated	Direct	Specify a 25 mm raised-letter metal FDC sign	Verified
913.1	Fire-pump standard	Where provided, install fire pumps per this section and NFPA 20	Fire pump provided	Exception: 13D / SBC 1101–1102 pumps	Direct	A high-rise 13 system will typically need a pump; design to NFPA 20 (not imported)	External verification
913.2.1	Fire-pump room rating	Locate fire pumps in rooms separated by 2-hour fire barriers or 2-hour horizontal assemblies, or both	Fire pump in the building	Exception 1 (1-hour in a fully sprinkled non-high-rise) does not apply to high-rise. Exception 2: physical separation per NFPA 20	Direct	Enclose the pump room 2 hours ; do not take the 1-hour non-high-rise exception	Verified
913.2.2	Fire-pump circuit survivability	Protect cables for survivability of circuits supplying fire pumps by UL 2196 1-hour listed cable, a 1-hour electrical circuit protective system, 1-hour construction, or encasement in not less than 50 mm of concrete	Fire-pump supply circuits	Exception: cables inside a fire-pump or generator room that is already fire-resistance separated	Direct	Show a 1-hour survivability method from the source to the pump room	Verified
913.3	Pump-room temperature	Maintain the pump room or pump house above 5 °C	Pump room/house where required	Engine manufacturer’s minimum may be higher (913.3.1)	Direct	Heat the pump room above 5 °C (and above 4 °C per 902.1.3)	Verified
914.1.1–914.1.2	Shaftway markings	Mark outside openings onto a multi-story hoistway/shaftway, and interior door/window openings onto a shaftway, with SHAFTWAY in red letters not less than 150 mm high on a white background	Openings onto communicating shafts	Interior markings not required where the opening is readily discernible as a shaftway by construction	Direct	Put 150 mm SHAFTWAY signs on exterior shaft openings and on non-obvious interior shaft doors	Verified

18. Carbon monoxide detection and emergency responder radio

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
915.1.1	Group R CO detection	Provide carbon monoxide detection in Group R occupancies in 915.2 locations where any 915.1.2–915.1.6 condition exists	Group R with a listed fuel-burning or garage condition	Open/enclosed parking garages per 406.5/406.6 are not “private garages” (915.1.6)	Conditional	If the tower is all-electric with no fireplace, furnace or attached private garage, CO detection is not charged by this section	Verified
915.1.2–915.1.4	Fuel-burning CO triggers	Detect in units that contain a fuel-burning appliance or fireplace; units served by a fuel-burning forced-air furnace; and units in buildings that contain fuel-burning appliances/fireplaces	Those fuel-burning conditions	915.1.3: detector in the first room served by each main duct with alarm to an approved location. 915.1.4: no communicating openings, or listed remote-detector locations	Conditional	Map every gas appliance; apply the communicating-opening exceptions only where proven	Verified
915.1.5	Attached private-garage CO	Provide CO detection in units in buildings with attached private garages	Attached private garage	No communicating openings; units more than one story above or below; open-ended-corridor connection; detector between garage openings and units. 915.1.6: 406.5/406.6 parking is not a private garage	Conditional	Typical podium parking is 406.5/406.6, not a private garage; apply only to true attached private garages	Verified
915.2.1	Dwelling-unit CO location	Install outside each separate sleeping area in the immediate vicinity of the bedrooms; if a fuel-burning appliance is in a bedroom or its attached bathroom, install within the bedroom	Required dwelling-unit CO	Sleeping-unit path is 915.2.2	Conditional	Place CO alarms at bedroom halls, and in bedrooms that contain fuel-burning equipment	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
915.4.1	CO-alarm power	Primary power from commercial building wiring; battery on primary-power loss; permanent wiring without a disconnect other than overcurrent protection	CO alarms (UL 2034)	Buildings without commercial power may use battery-only alarms	Conditional	Circuit CO alarms like smoke alarms	Verified
918.1	Emergency responder radio	Provide emergency responder radio coverage in all new buildings in accordance with SBC 801 Section 510	New buildings	Wired 907.2.13.2 substitute only where approved in lieu of SBC 801 §510	Direct	Design in-building radio coverage to SBC 801 §510; do not import those signal-strength values here	External verification

19. Project-use controls

1. Use **Verified** rows for initial life-safety and fire-protection scoping after the row trigger and branch are confirmed.
2. Treat every **Verify source** row (flattened extinguisher tables, visible-alarm Table 907.5.2.3.2, 907.6.6 OCR, missing 903.2.11.2 heading, cut-off 909/910 equations) as a design hold point. No affected value is to be placed in issued-for-approval documents without a published-source check.
3. Do not specify NFPA 13R for this high-rise. Do not take 13-only code reductions on any residual 13R or 13D hypothesis.
4. Do not import 403.3, 1023.12, ICC A117.1, SBC 501, SBC 801 §510, NFPA 13/14/20/72 or commentary **18.35 m²** FCC area.
5. Do not apply Group A/E/F/H/I/M/S occupancy-threshold rows, mall/stage/amusement rules, or Table 907.5.2.3.2 R-1 visible counts to this R-2 tower unless the gap register already opened that use.
6. Record sprinkler standard, FD-access datum, mixed-use, FCC location and SCD NOC decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

20. Coverage summary

Internal inventory of the attached Chapter 9 extract (numbered code, exceptions, tables, footnotes; commentary and figures excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 310
- **Verified:** 251
- **Verify source:** 59

Counts by top-level section

Section	Records
901	2
902	3
903	76
904	12
905	22
906	37
907	76
908	0
909	40
910	10
911	6
912	9
913	5
914	4
915	3
916	4
917	1
918	0

Appended-table coverage

Appended table	Records	Verify source records
Table 903.2.5.2 (H-5 hazard names; no SI cells)	0	0
Table 903.2.11.6 (outbound section pointer)	0	pointer OCR hold (non-numeric)
Table 906.1 (SBC 801 pointer)	0	pointer OCR hold (non-numeric)
Table 906.3(1) Class A distribution	12	12
Table 906.3(1) footnotes a, c	4	0
Table 906.3(2) Class B distribution	10	10

Appended table	Records	Verify source records
Table 907.5.2.3.2 visible alarms (I-1/R-1)	34	34

Coverage cross-check against SBC 201 Chapter 9 Fire Protection Systems (2024)_CS.md was topics-only: Group R sprinkler-throughout; 13R four-story / **9 m** limit; high-rise EVACS/FCC; standpipe height. No CS.md value was copied into a matrix cell.

21. Unresolved-source register

Hold points for the 59 **Verify source** inventory records, plus two non-numeric OCR/heading holds (903.2.11.2, 907.6.6). Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 903.2.11.6	Concatenated section/subject pointer, including 403.3 High-rise buildings	Record the outbound pointer only; obtain published Chapter 4 / 403.3 before using high-rise suppression text
Table 906.1	Concatenated SBC 801 section list	Do not reconstruct extinguisher locations from flattened 801 cites
Table 906.3(1)	12 concatenated Light/Ordinary/Extra cells (2-A / 4-A, 280 / 140 / 100 m², 1,045 m², 23 m)	Verify the published table before locking Class A rating and travel; footnotes a/c (9.5 Liters , 1-A/2-A/4-A equivalents) are readable
Table 906.3(2)	Extra-High row concatenated 20-B1540-B980-B15 ; remaining cells flattened	Do not reconstruct 20-B / 40-B / 80-B travel; verify the published table if a Class B hazard exists
Table 907.5.2.3.2	Concatenated I-1/R-1 visible-alarm counts	Omit from R-2 project-use; do not use reconstructed sleeping-unit percentages
903.2.11.2	Code heading missing; chute-sprinkler body is attached after 903.2.11.1.3	Confirm published heading and alternate-floor / lowest-intake placement before issuing the chute detail
907.6.6	Source text OCR-repeats "monitored by this chapter or by SBC 801"	Use 901.6.3 for supervising-station intent; verify published 907.6.6 wording
909.6.2	Door-opening-force equation cut off after "determined by:"	Do not invent the force equation; use 909.20.5 / 909.20.6.2 numeric caps where those methods apply
909.10.1	Exhaust-fan temperature-rise equation cut off after "computed by:"	Do not invent the temperature-rise formula; keep fans as 909.10 performance
910.3.3	Smoke-and-heat vent area formula cut off after "calculated as follows:"	Group F-1/S-1 vents are not typical; verify the published formula if 910 is opened